

Retrieval-Augmented and Data-Enhanced Stock Market Prediction with Sentiment Integration and Data Augmentation for Crash Resilience

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Abstract— Financial markets are influenced by fast-changing news events and technical indicators, where every decision directly affects returns and risk. This study introduces a trading framework that integrates multi-source financial news with technical analysis to generate real-time signals. A Retrieval-Augmented Generation (RAG) module is employed to filter and prioritize stock-specific updates using Named Entity Recognition and sentiment scoring, ensuring that only impactful events shape trading outcomes. This design supports intraday, swing, and positional strategies, while a dynamic risk layer continuously adapts stop-loss and exit points to reflect cautious, human-like behavior under stress. The crash detection ability is done by analyzing international news by allowing the system to check early signs of major disruptions such as policy shifts, conflicts, or economic downturns. This framework connects to broker platforms like Zerodha Kite and Alpaca by Python through FastAPI and uses Streamlit for real-time monitoring. Logging tools are used to record trade performance, and sandbox environments and backtesting offer transparent assessment and validation based on historical data. **Index Terms**— Artificial Intelligence, Automated Trading, Sentiment Analysis, Financial News, Retrieval-Augmented Generation (RAG), Named Entity Recognition (NER), Technical Indicators (SMA, EMA, RSI, MACD, Bollinger Bands), Risk Management, Crash Detection, Emotional Trading Strategy, FinBERT, Real-Time Stock Prediction, Backtesting, Zerodha Kite, Alpaca API, Streamlit, FastAPI, Data Augmentation.

I. INTRODUCTION

Stock markets are influenced not only by price variations but also by ongoing financial news, global events, and macroeconomic changes. Traders rely on news analysis, technical charts, and patterns to help them make decisions, but because information moves so quickly, there is frequently little time for prompt execution. Investors are exposed to greater risk and lost opportunities as a result of this gap. Traditional algorithmic systems attempt to reduce the burden through automation, yet most rely on either technical indicators or sentiment signals in isolation. These single-source approaches

might work well in periods of stability, but they usually fall short in volatile times when price behaviour and contextual news work together to shape market outcomes. As a result, investors are still vulnerable to shocks brought on by shifts in policy, company results, or geopolitical unrest. To overcome these limitations, this research proposes a trading framework that integrates multiple sources of information. Financial news is collected from reliable platforms such as Moneycontrol, NDTV, and Economic Times. Named Entity Recognition (NER) filters unwanted data, ensuring only stock-specific updates are processed. After that, FinBERT is used to assign sentiment polarity, and indicators like the Simple Moving Average (SMA), Exponential Moving Average (EMA), Relative Strength Index (RSI), and Moving Average Convergence Divergence (MACD) are used to cross-validate the signals. Retrieval-Augmented Generation (RAG), a key component of the framework, facilitates contextual decision-making by mixing historical scenarios with current events. Improved entry and exit tactics, dynamic stop-loss adjustment, and increased resistance to disruptive circumstances are all made possible by this mechanism. Global news augmentation and volatility monitoring improve crash detection even more by enabling early identification of destabilising events. Before going live, additional modules for trade logging, sandbox testing, and backtesting offer transparency and validation. By combining technical indicators with sentiment-driven insights, the proposed framework delivers a scalable, adaptive, and risk-aware trading assistant. It aims to reduce decision-making pressure while improving investor readiness in both routine and uncertain market environments.

II. RELATED WORKS

In [1], Araci introduced FinBERT, a domain-specialized language model for the financial domain that outperformed non-domain-specialized NLP models by an overwhelming wider margin in sentiment classification accuracy, demonstrating the necessity of domain adaptation for financial text processing. In [2], Lewis et al. introduced Retrieval-Augmented Generation (RAG), a system that combines information retrieval with large language models to provide factual grounding and contextual reasoning. This can be applied as is to financial robots in terms of real-time precision and good decision-making. Ding et al. in [3] proposed an

event-driven stock forecasting method by deep learning with focus on how short- and long-term behavior of stocks is governed by finance events present in news. In [4], Zhou et al. propose a two-stage corporate event detection model that can identify high- and low-level financial news events and therefore enable the prediction of sudden development-led mispricing of stocks.

In [5], Zhai et al. combined technical indicators and news sentiment to predict day-to-day movement in stock price and emphasized how media news significantly influences investor sentiment regardless of conventional chart-based indications. To enhance prediction robustness, Wen et al. in [6] researched augmentation techniques for financial time-series and demonstrated how the noise injection regimes and time warping reduce overfitting as well as improve generalizability. Digging further into the prediction of crashes, Chen et al. in [7] introduced the Crisis-Alert system using Random Forest and XGBoost on high-frequency data to predict stock market crashes and demonstrated that machine learning can do very well in crisis signal detection.

Shao, in [8], contrasted the influence of sentiment expressed in financial news on stock performance and concluded that negative sentiment placed greater impacts in high-technology industries compared to conventional industries. According to this, Amola in [9] developed an NLP-founded financial news sentiment model with positive, negative, or neutral classification having greater market predictive ability when combined with quantitative information. Previously, Tetlock in [10] analyzed Wall Street Journal newspaper article media pessimism and determined that tone of pessimism to predict short-term stock price falls as well as abnormal trading volume.

Gu et al. in [11] also recently presented a FinBERT-LSTM hybrid model that combines the strengths of sentiment analysis with sequential time-series modeling and improved prediction performance on NASDAQ-100 datasets over baseline models. Finally, Xiao et al. in [12] proposed FinSrag, a retrieval-augmented large finance language model architecture, that better predicted stock prices by extrapolating real-time market news into historical trading dynamics. These works establish the validity of sentiment, event, augmentation, and retrieval-based reasoning to perform precise and adaptive financial forecasting.

III. METHODOLOGY

This trading bot is implemented in Python (Visual Studio Code) with deployment on Streamlit, and API services handled using FastAPI. All source code is maintained on GitHub. Unlike traditional bots, this system uses Retrieval-Augmented Generation (RAG), allowing decisions to be informed by both real-time news and historical trading patterns, while also integrating technical analysis, sentiment, and crash detection.

A . Data Collection and Visualization

This bot retrieves Market Data using yFinance, covering OHLCV (Open, High, Low, Close, Volume) for the Indian Stocks like BEL, SI, IRCTC, Indigo, TATA motors etc. Users can select a time frame from 1 Day to 1 Year. We have used plots for interactive candlestick and line charts, enabling detailed study of

market trends and volatility.

Candlestick Chart



Figure 1 : Candlestick Graph

B . Financial News and Sentiment Processing

Real-time stock market news is continuously collected from multiple APIs, including Moneycontrol, NDTV, and Economic Times. Named Entity Recognition (NER) filters relevant content for each company or sector to guarantee that only specific stock updates are kept. After processing the articles. FinBERT, which assigns a confidence score to each sentiment polarity classification that is positive, negative, or neutral . Because high-impact sentiment is given priority in the decision-making process, trading strategies are kept flexible and responsive especially during important policy announcements, earnings reports, global market swings, or geopolitical events.

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Figure 2 : News Sentiment and Trading Action Visualization

C. Technical Indicators and Trading Logic

The framework incorporates standard technical indicators such as Bollinger Bands, Moving Average Convergence Divergence (MACD), Relative Strength Index (RSI), Simple Moving Average (SMA), and Exponential Moving Average (EMA). Sentiment analysis is then used to cross-validate the buy, hold, or sell signals produced by these indicators. To guarantee alignment with actual market conditions, technical outputs may be overridden in situations where strong sentiment signals are identified.

D. Risk Management and Emotional Strategy

This framework includes daily drawdown limits, capital allocation plans, and adjustable stop-loss mechanisms to reduce risk. An emotional strategy layer is incorporated in addition to fixed rule-based controls to simulate human-like caution during volatile times. The system maintains disciplined trading behaviour by automatically lowering exposure, tightening stop-loss thresholds, and avoiding aggressive entries when it detects sudden market shifts.

E. Retrieval-Augmented Generation (RAG) Core

The RAG module is the decision-making hub of the system. It performs three key tasks:

- **Data Retrieval:** RAG Module collects stock-specific news, filtered NER outputs, technical indicator signals, and sentiment scores..
- **Contextual Analysis:** RAG Module matches current events with historical scenarios (e.g., government contract announcements for BEL or policy changes for SBI) to evaluate possible market outcomes..
- **Decision Prompting:** After integrating data, the RAG will generate a structured trading decision (buy, hold, or sell) tailored to intraday, swing, or positional trading.

These are then relayed to broker Application Programming Interfaces (APIs) such as Zerodha Kite, Alpaca, Fyers, and Alice Blue for virtual or real transactions. Traditional systems respond to standalone indicators in contrast to, the Retrieval-Augmented Generation (RAG) unit combines disjointed streams of data to produce context-dependent decision-making. This enables the system to emulate human-like reasoning through the availability of sentiment, technical indicators, and event-driven data in an attempt to reduce over-response vulnerability against single-market inputs.

```
{
  "ticker": "BEL",
  "sentiment": "neutral",
  "event": "Nifty extends winning streak as IT stocks rally, banking sector lags",
  "confidence": 0.53,
  "recommended_action": "HOLD",
  "stop_loss": 1.34,
  "take_profit": 3.79
}
```

Figure 3 : JSON Representation of Sentiment and Trading Action

F. Crash Detection with Data Augmentation

Crash detection is supported by:

- **Tracking volatility for sudden falls ($\geq 3-5\%$) and increase in volumes.**
- Data augmentation techniques such as time warping and noise injection to attempt to expose the model to abnormal crash scenarios.
- Event-news monitoring where geopolitical shocks or events occurring locally are caught by NER, activating defensive trading mode.
- During a crash, the bot becomes capital-sensitive by closing positions and reducing over-aggressive trading.

G. Backtesting and Sandbox Testing

This bot enjoys the privilege of a sandbox mode using Backtrader to test against historical data sets before live deployment. Sharpe Ratio, Win Rate, and Maximum Drawdown are calculated for use in consideration for estimating the power of resilience in the context of performance.

H. Backtesting and Sandbox Testing

Sentiment-driven logic, performance match results, and entry and exit point data are all used to track every trade. Additionally, users can create a personal watchlist, dynamically filter stocks according to trading volume, sentiment indicators, or RSI values, and even get real-time alerts via email or Telegram. Furthermore, the framework facilitates extensive sandboxing and backtesting, allowing strategies to be tested on historical data using metrics like Sharpe ratio, win ratio, and drawdown limit. An additional factor contributing to credibility is the increased confidence, risk sensitivity, and sensitivity of a two-lane logging and testing framework to investors regarding the performance of strategies under different market conditions.

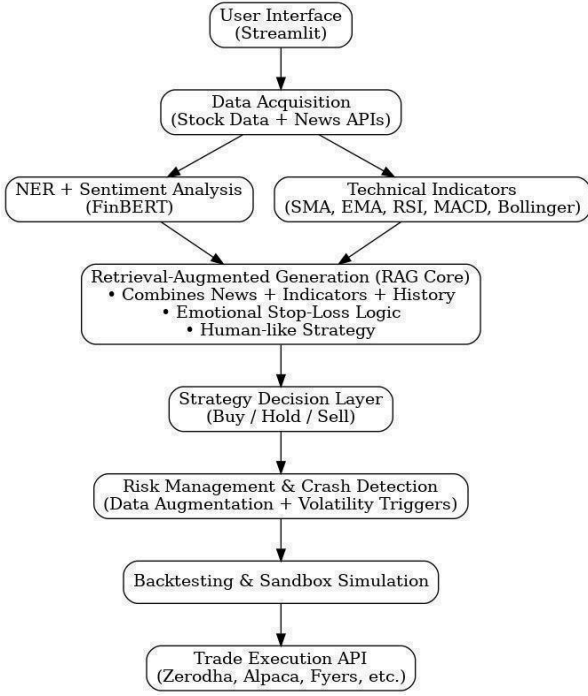


Figure 4: Flowchart of the Proposed Trading Framework

IV. FUTURE WORKS

Despite the integration of real-time sentiment analysis, technical indicators, crash warnings, and Retrieval-Augmented Generation (RAG) into the decision in the presented system, a lot of scope remains for further enhancing it. Increased adaptability, robustness, and investor-friendliness are predicted for future expansions to make the system more usable and accessible.

A. Dynamic Stop-Loss and Emotional Strategies

At the moment, our bot has a stop-loss module which has fixed levels. We will be adding adaptive stop-loss as an automatic reaction based on volatility, strength of sentiment, and historical price action in later versions. An emotional strategy layer will also be added in a way that the bot will respond wisely during periods of volatility like an experienced trader.

B. Long-Term Investment and Mutual Fund Support

The present version is designed to support intraday, swing, and positional strategies. Long-term investment attributes will be provided in subsequent releases based on studies of the elementary metrics such as EPS growth, dividend trend, and sector performance. Systematic Investment Plan (SIP) and mutual fund facility will also be provided, which will render the system beneficial for small investors looking for safety first and creating long-term wealth.

C. Stock-Specific Behavioral Modeling

One of the drawbacks of most AI trading sites is to treat stocks equally with equal logic. Each stock in real life has its own

personality for example, BEL can vigorously counter defense orders, and SBI can counter policy or banking reforms. Future models will be able to comprehend these unique stock "personalities," so strategy will be fitting for each asset's unique quirks.

D. Smarter News Intelligence and Multi-Source Sentiment

Although the system currently extracts opinion using FinBERT and Named Entity Recognition (NER), future ones will be accessing opinion from an array of sources like Twitter, Reddit, and sophisticated breakdowns of financial reports within the space. By leveraging state-of-the-art Natural Language Processing (NLP) pipelines, the platform will not only identify sentiment polarity but also deduce event context, censor objectionable noise, and overall trade accuracy.

E. Cross Market and Crash Resilience

The crash detection system would be further improved by high-intensity stress testing across vast regimes of regimes of the market, i.e., bull regime, bear regime, sideways regime, and volatile regime. Time warping, price shock, and SMOTE-based methods towards crash simulation would be employed in training the system for early warning sign identification. This would make the system robust to highly extreme market regimes and reduce the vulnerability of the system to abrupt shocks.

F. Explainable AI (XAI) Integration

To make the investors more confident in the future, future releases will add explainability to the decision pipeline. The system rather than providing buy or sell will provide context-relevant reasons such as: "Trade initiated due to $RSI < 30$ and negative news sentiment around earnings." Human-readable reasons will enhance transparency, comprehension of the automaton, and trust establishment in the system suggestions.

V. CONCLUSION

The present research gives a latest artificial intelligence (AI) trading model that is customized for the Indian stock market. The model differs from those constructed using discrete signals in the sense that the model is given context-based financial data and technical signals for supporting risk-averse decisions in managed environments. The system is robust with improved news filtering, adaptive stop-loss management, and crash protection capabilities to balance fear on the part of investors without losing capital during a market crash. Furthermore, emotion-sensitive trading regulations encourage discipline in cases of utmost uncertainty without losing the capacity to pursue intraday, swing, and position opportunities. Overall, the architecture is capable of potential extension to long-term investment plans like systematic investment schemes (SIPS) and mutual fund management. Therefore, the structure provides the investor with an open, stable, and reactive trading counterparty that brokers profitability against capital protection.

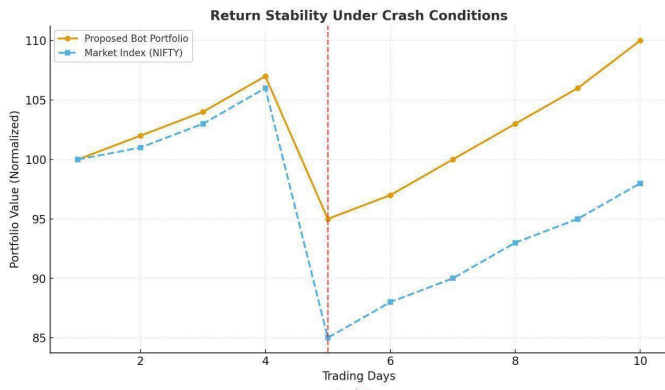


Figure 5: Return Stability of Proposed Trading Bot Compared to Market Index During Crash Conditions

This figure depicts the comparative performance of the suggested bot portfolio when compared with the market index (NIFTY) under crash scenarios. While the index takes time to recover from the crash, the suggested system shows resistance in lowering drawdowns, recovering faster, and maintaining positive momentum all along. It is because of this reason that the system is capable of providing stability, holding capital, and providing uniform returns even amidst challenges, thereby making it a suitable, human-like trading companion in situations of financial stress

IV. REFERENCES

- [1] D. Araci, "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models," arXiv preprint arXiv:1908.10063, 2019. [Online]. Available: <https://arxiv.org/abs/1908.10063>
- [2] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, Y. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Advances in Neural Information Processing Systems (NeurIPS), 2020. [Online]. Available: <https://arxiv.org/abs/2005.11401>
- [3] X. Ding, Y. Zhang, T. Liu, and J. Duan, "Deep Learning for Event-Driven Stock Prediction," in Proc. 24th Int. Joint Conf. Artificial Intelligence (IJCAI), Buenos Aires, Argentina, 2015, pp. 2327–2333. [Online]. Available: <https://dl.acm.org/doi/10.5555/2832415.2832572>
- [4] Z. Zhou, L. Ma, and H. Liu, "Trade the Event: Corporate Events Detection for News-Based Event-Driven Trading," Findings of the Association for Computational Linguistics (ACL-IJCNLP), 2021, pp. 2114–2124. [Online]. Available: <https://aclanthology.org/2021.findings-acl.186/>
- [5] M. Nguyen and Q. Le, "Combining News and Technical Indicators in Daily Stock Price Trends Prediction," in Proc. Int. Conf. on Knowledge and Systems Engineering (KSE), 2015, pp. 135–140. doi: 10.1109/KSE.2015.7320393
- [6] E. Iwana and S. Uchida, "Time-Series Data Augmentation for Deep Learning: A Survey," arXiv preprint arXiv:2002.12478, 2023. [Online]. Available: <https://arxiv.org/abs/2002.12478>
- [7] Y. Chen, X. Andrew, and S. Supasanya, "CRISIS ALERT: Forecasting Stock Market Crisis Events Using Machine Learning Methods," arXiv preprint arXiv:2401.06172, 2024. [Online]. Available: <https://arxiv.org/abs/2401.06172>
- [8] M. Garcia, "The Impact of News Media Sentiment on Financial Markets," SSRN Electronic Journal, 2013.

[Online] Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2290646

- [9] M. Amola, "Sentiment analysis in financial news: Enhancing predictive models for stock market behavior," *ResearchGate*, Mar. 2025. [Online]. Available: https://www.researchgate.net/publication/390056578_Sentiment_Analysis_in_Financial_News_Enhancing_Predictive_Models_for_Stock_Market_Behavior
- [10] W. Gu, Y. Zhong, S. Li, C. Wei, L. Dong, Z. Wang, and C. Yan, "Predicting Stock Prices with FinBERT-LSTM: Integrating News Sentiment Analysis," in Proc. 2024 8th Int. Conf. Cloud Big Data Comput., 2024. [Online]. Available: <https://dl.acm.org/doi/full/10.1145/3694860.3694870>
- [11] M. Xiao, Z. Jiang, L. Qian, Z. Chen, Y. He, Y. Xu, Y. Jiang, D. Li, R.-L. Weng, M. Peng, J. Huang, S. Ananiadou, and Q. Xie, "Enhancing Financial Time-Series Forecasting with Retrieval-Augmented Large Language Models," arXiv preprint arXiv:2502.05878, 2025. [Online]. Available: <https://arxiv.org/abs/2502.05878>
- [12] A. Karasan, Ö. S. Alp, and G.-W. Weber, "Machine learning approach to stock price crash risk," *Annals of Operations Research*, vol. 350, pp. 1053–1074, 2025. [Online]. Available: <https://link.springer.com/article/10.1007/s10479-025-06596-7>
- [13] "Technical Patterns and News Sentiment in Stock Markets," *Journal of Finance and Data Science*, vol. 10, Dec. 2024, Art. no. 100145. [Online]. Available: <https://doi.org/10.1016/j.jfds.2024.100145>
- [14] J. Deveikyte, H. Geman, C. Piccari, and A. Provetti, "A Sentiment Analysis Approach to the Prediction of Market Volatility," arXiv preprint arXiv:2012.05906, 2020. [Online]. Available: <https://arxiv.org/abs/2012.05906>
- [15] S. Yoon and H. Takahashi, "Intraday Trading Dynamics of Characteristics and Sentiment Tendencies of Past News in the Tokyo Stock Exchange Market," *Computational Economics*, vol. 65, no. 1, Jan. 2025. [Online]. Available: <https://link.springer.com/article/10.1007/s10614-024-10768-1>

